

Section 1: Reading and Writing

Section 1, Module 1: Reading and Writing

27 Questions

Question 1

Forward-thinking plumbing design is _____ the possibility that a building's needs will change over time in unpredictable ways. This foresight manifests in oversized main lines, accessible junction points, and strategically placed valves that will allow future modifications without requiring walls and foundations to be extensively opened.

Which choice completes the text with the most logical and precise word or phrase?

- A. substantiates
- B. quantifies
- C. accommodates
- D. transcends

Question 2

When building outdoor structures like decks or fences, carpenters must carefully _____ lumber options, selecting naturally rot-resistant woods like cedar or pressure-treated pine that can withstand moisture exposure, as conventional untreated wood would quickly deteriorate when subjected to rain, humidity, and ground contact.

Which choice completes the text with the most logical and precise word or phrase?

- A. offset
- B. evaluate
- C. conceal
- D. sustain

Question 3

Though most hoaxes perpetrated as jokes by mischievous users of Wikipedia, an online encyclopedia that almost anyone can freely edit, have quickly been detected and removed, a few fictitious entries, such as those for the American new wave band Transformatix and the 19th-century German poet Emilia Dering, persisted on the site for many years before they were finally recognized as _____ and deleted.

Which choice completes the text with the most logical and precise word or phrase?

- A. revelations
- B. analyses
- C. enhancements
- D. pranks

Question 4

Whatever the general attitude toward William Greaves's 1968 film *Symbiopsychotaxiplasm: Take One* and Spike Lee's 1992 film *Malcolm X* when they were initially released, both films now tend to be regarded quite _____. In 2018, for example, critics for the New York Times praised the former as "a vital artifact of its time" and the latter as "electrifying."

Which choice completes the text with the most logical and precise word or phrase?

- A. skeptically
- B. neutrally
- C. favorably
- D. strangely

Question 5

Software that automatically detects suspicious behavior on computer networks has become so sensitive that much of the activity it detects is _____, which means that rather than replacing human experts, such software has increased the need for cybersecurity specialists to interpret the flagged activity and determine whether it is a cause for concern.

Which choice completes the text with the most logical and precise word or phrase?

- A. invariable
- B. imperceptible
- C. malicious
- D. ambiguous

Question 6

A student is writing a research paper on the history of irrigation in the southwestern United States, situating the development of the Salt Spring Valley Reservoir (a man-made body of water created in Calaveras County, California, in 1882) in a larger historical context. The student claims that some modern-day irrigation methods used in southwestern states, such as California, Arizona, and New Mexico, have a lineage that long predates the completion of the Salt Spring Valley Reservoir.

Which quotation from a historian best supports the student's claim?

- A. "In the decades following the Second World War, irrigation of arid areas for agriculture increased to such a degree that it now accounts for roughly 70% of the world's water usage."
- B. "The importance of the development of irrigation infrastructure in the American Southwest today cannot be overstated, since it is the most common means of conveying water for food production."
- C. "The irrigation system developed by the Hohokam people in what is now central Arizona in the 7th century CE was simple but made use of hydraulic engineering design principles that are still utilized by today's engineers."
- D. "Sprinkler irrigation systems, which were developed in the late 20th century, are a convenient technique of irrigating that makes use of machinery in order to spray water in all directions. Many areas of the Southwestern United States are currently irrigated by this method."

Question 7

New York-based artist Uman is no stranger to crossing borders: born in Somalia, she lived in Kenya and Denmark before moving to the United States. Her art, too, straddles multiple boundaries. Drawing on knowledge of everything from East African fabrics to Arabic calligraphy, Uman's kaleidoscopic paintings often combine abstract shapes with anthropomorphic elements, like eyes, mouths, and gestures. Moreover, she seeks to cultivate a practice of "constant evolution"—unaltered from any prescriptive agenda, so that her art navigates a liminal space between tradition and autobiography, community and itinerancy, and activism and contemplation.

Which statement about Uman's art is most strongly supported by the text?

- A. Its spontaneity makes it challenging for critics to recognize the conventions of genre that Uman chooses to observe.
- B. Its rejection of established aesthetic criteria can be traced to Uman's ambivalence toward cultural traditions.
- C. Its adherence to certain dichotomies sets it apart from the work of most abstract artists based in the United States.
- D. Its openness to experimentation is evident in its diverse influences and refusal to adhere to a single conceptual frame.

Question 8

The following text is adapted from Rupa Bajwa's 2004 novel *The Sari Shop*. Ramchand works as an assistant in a shop in Amritsar, India.

Even at ten in the morning, the bazaar was throbbing with activity. The halwai [baker] was already installed in front of the Mishthan Sweet Shop, pressing jalabi batter into squiggly shapes that floated and simmered in the oil in a big iron cauldron. All the shops had opened for the day and, Ramchand noted guiltily, all the shop assistants were already in place, trying to sell things with fixed, attentive smiles on their shiny, bathed faces.

Based on the text, what can most reasonably be inferred about Ramchand?

- A. Ramchand is overwhelmed by a bazaar after seeing the chaos of other shop assistants selling goods.
- B. Ramchand is enthusiastic about being at work after seeing shop assistants who are excited about their jobs.
- C. Ramchand is remorseful about arriving to work later than other shop assistants he sees.
- D. Ramchand is fascinated by frying jalabi batter while wandering to work as a shop assistant.

Question 9

Devon A. Mihesuah is a novelist and a citizen of the Choctaw Nation of Oklahoma. Critics and fellow writers have praised her work for combining traditional Choctaw stories with popular genres of literature, such as mystery fiction. But Mihesuah is not the only Indigenous writer to blend traditional stories with popular literature. In the 2020 novel *The Only Good Indians*, Stephen Graham Jones successfully blended the oral storytelling tradition of his tribe, the Blackfeet Nation, with horror fiction.

Which choice best states the main idea of the text?

- A. Recent Indigenous fiction writers are mixing traditional stories with popular literary genres.
- B. Several famous Indigenous writers increased their book sales when they began incorporating traditional stories.
- C. Non-Indigenous authors should consider the expectations of Indigenous readers.
- D. The success of Devon A. Mihesuah has influenced non-Indigenous authors who write in popular literary genres.

Question 10

Betty Smith's 1943 novel *A Tree Grows in Brooklyn* is regularly described as autobiographical. That characterization is apt—there are many parallels between the experiences of the novel's protagonist, Francie Nolan, and those of Smith—but it should not be taken to mean that every person or event depicted in *A Tree Grows in Brooklyn* has a real-life analogue. Much of the novel is pure invention, and readers who neglect this fact and instead focus excessively on correspondences between the novel and Smith's life can thus _____.

Which choice most logically completes the text?

- A. misrepresent *A Tree Grows in Brooklyn* as being more widely read than it actually is.
- B. fail to appreciate how much creativity Smith shows in the book.
- C. overemphasize the extent to which Smith took inspiration from earlier writers.
- D. refuse to acknowledge that Smith drew on real-world material when writing *A Tree Grows in Brooklyn*.

Question 11

In 2002, Richard Long created *Macduff Circle*. In this work, Long set up a large circle made out of a balanced array of different-sized pieces of slate. Long's work is part of the land art movement, in which artists reject the confinement and commercialization of art galleries. These artists usually install large-scale pieces outdoors, often in surprising locations.

Based on the text, what can be inferred about *Macduff Circle*?

- A. It may have had more commercial success in a different outdoor location.
- B. It marked a shift in the popularity of the land art movement.
- C. It was created early in Long's career.
- D. Its location may be considered surprising.

Question 12

A student is writing an essay on the subject of cultured meat, which is grown in a laboratory and is intended to help reduce the number of livestock harvested for food. The student wants to make the case that cultured meat production may be useful in responding to an expected widespread change in the public's behavior.

Which quotation from a publication by a researcher would most effectively support the student's claim?

- A. "Researchers who advocate for the development of lab-grown meat claim that it's better for the environment than conventional meat because it requires less water consumption."
- B. "The flavor of meat from livestock differs across species (from pig to chicken to cow), and is also influenced by farming conditions and the breeds and genders of animals."
- C. "A growing population that is including significantly more meat in its diet will contribute to an increasing demand for meat products in the 21st century."
- D. "Consumers tend to believe that using less packaging when selling meat products in stores would have a significant effect on the environment."

Question 13

In 2005, Swedish researchers studying flawed choice recall devised an experiment in which participants were shown two photos of different faces and asked to select the one they found more attractive. The researchers appeared to give the participants the photo they had chosen, but in fact used sleight of hand to give them the photo of the person they had not chosen. Most participants accepted the swap and, when asked to explain their initial preference, gave justifications for having chosen the face in the photo they were given. Subsequent researchers have replicated this finding with other choices, including jam preferences.

Based on the text, why was the sleight-of-hand step an important part of the experiment?

- A. It allowed the researchers to control for the possibility that some participants found the two faces roughly equally attractive.
- B. It enabled the researchers to determine whether the thought processes involved in selecting which photo had the more attractive face was similar to the thought processes involved in selecting a preferred jam.
- C. It helped the researchers test whether participants would embrace an outcome they thought was consistent with their own choices even when it was truly contradictory.
- D. It allowed the researchers to measure the willingness of participants to change their positions when presented with persuasive counterarguments.

Question 14

Positive expressive writing is self-reflective writing aimed at exploring positive topics and feelings like hopefulness. Techniques of positive expressive writing include one called “benefit finding,” which involves writing about positives that come from difficult experiences, and “satisfaction processes,” which involves writing in-depth about how certain experiences lead to feelings of fulfillment. In a large-scale analysis, Lauren Hoult and colleagues determined that while different techniques produce varying outcomes, they all tend to have a more consistent effect on well-being and positive outlook than on physical health and stress.

Which choice best describes the overall structure of the text?

- A. It introduces a study about positive expressive writing and then explains that the researchers are planning a follow-up study.
- B. It gives an overview of positive expressive writing and then states a research finding about this type of writing.
- C. It identifies several positive emotions and then asserts that positive expressive writing should be done daily to amplify those emotions.
- D. It describes one popular technique of positive expressive writing and then names the techniques researchers have found to be most useful.

Question 15

Cane is a 1923 novel by Jean Toomer. In the novel, Toomer mentions a road in rural Georgia called Dixie Pike and describes it as having a deep connection to a faraway place, writing, _____.

Which quotation from *Cane* most effectively illustrates the claim?

- A. “And when the wind is from the South, soil of my homeland falls like a fertile shower upon the lean streets of [Washington, DC].”
- B. “The Dixie Pike has grown from a goat path in Africa.”
- C. “If it were dusk, then [Fern’s eyes would] wait for the search-light of the evening train which you could see miles up the track before it flared across the Dixie Pike, close to her home.”
- D. “I leave the men around the stove to follow [Carma] with my eyes down the red dust road.”

Question 16

A restaurant is installing window film to create a more comfortable indoor atmosphere. This is because during dinner service, the sunlight gets in people's eyes while they eat. The restaurant owner wants to reduce the glare but still let customers enjoy the sunset. To help meet these goals, a window-film technician recommends selecting a film that reduces glare as much as possible while still allowing more than 15% of visible light to pass through the windows.

Film name	Visible light allowed through	Glare reduction
No film used	90%	14%
GlowGuard	33%	52%
ShieldView	22%	65%
BronzeShine	16%	77%
ShadeBlock	7%	82%

Based on the window-film technician's recommendation and the data in the table, which film should the restaurant owner choose?

- A. BronzeShine
- B. ShieldView
- C. ShadeBlock
- D. GlowGuard

Question 17

When studying the concrete used to build ancient Roman structures, MIT researcher Admir Masic analyzed small white mineral features, otherwise known as lime clasts, embedded in the concrete. _____ analysis led Masic to determine the lime clasts, which previous researchers had dismissed as mixing errors, were evidence of an intentional “hot mixing” process that gives the concrete remarkable self-healing properties.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. This
- B. All
- C. Those
- D. These

Question 18

Represented most commonly by a double vertical bar (||) or double oblique lines (/), _____ function as a deliberate break in the flow of poetry or music.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. the name comes from the Latin word for “cutting,” and the caesura's etymology reflects its
- B. the Latin word for “cutting” gives the caesura its name, an etymology that reflects the caesura's
- C. the caesura gets its name from the Latin word for “cutting,” an etymology that reflects its
- D. the etymology of the term “caesura” comes from the Latin word for “cutting,” reflecting its

Question 19

In their 2019 study of butterfly diversity at Rupa Lake in Nepal, researcher Bijaya Neupane and his colleagues counted one bluetail jester _____. The bluetail jester (*Symbrenthia niphanda*) belongs to the Nymphalidae family and is known to be very rare.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. butterfly and belonging to the Nymphalidae family.
- B. butterfly. Belonging to the Nymphalidae family,
- C. butterfly belonging to the Nymphalidae family.
- D. butterfly, belongs to the Nymphalidae family.

Question 20

In 1909, children in the US were introduced to Billy Possum, a stuffed toy opossum named after the newly elected president, William Taft, _____ by the entrepreneur Susie Wright Allgood. Billy Possum was promoted as a replacement for the popular teddy bear, which was named for the previous president, Theodore Roosevelt.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. Invent
- B. Is invented
- C. Invents
- D. Invented

Question 21

In 1840, Rowland Hill's penny postage system revolutionized mail in Britain, rendering obsolete previous systems in which senders paid variable rates based on distance traveled and recipients often refused delivery to avoid owing postage. Following _____ lead came similar systems implemented by reformers in other nations, such as Wilhelm Näff in Switzerland and José Clemente Pereira in Brazil.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. their
- B. there
- C. its
- D. it's

Question 22

Researchers' attempts to reintroduce the extinct toromiro tree to the Pacific island of Rapa Nui faced repeated failures despite the use of healthy seedlings and carefully chosen locations. _____ the researchers discovered that bacteria that the toromiro relied on to add nitrogen to the soil had also gone extinct, meaning that successful restoration would require reintroducing both the tree and its essential bacterial partners.

Which choice completes the text with the most logical transition?

- A. Most notably among these efforts,
- B. Similar to this project,
- C. Upon closer investigation,
- D. Despite these failures,

Question 23

Every US state has an associated state soil, which is typically selected by a group of experts, then passed through the state legislature to receive its official designation. For example, Nebraska's Holdrege soil was formally designated in 1979, and Michigan's Kalkaska soil in 1990. _____ years pass between a soil's selection and official designation, as the legislative process can be notoriously slow.

Which choice completes the text with the most logical transition?

- A. Similarly,
- B. In many cases,
- C. Therefore,
- D. Indeed,

Question 24

As anthropologist Cristina Grasseni explains, many artisanal cheeses in Italy, such as Robiola di Roccaverano from Piedmont and Casatella Trevigiana from Veneto, are highly valued for their regional authenticity: _____ they are prized for being made using local ingredients and methods typical of their respective geographic regions.

Which choice completes the text with the most logical transition?

- A. that is,
- B. then,
- C. rather,
- D. still,

Question 25

While researching a topic, a student has taken the following notes:

- Louis Ballard was a classical composer and citizen of the Quapaw Tribe.
- His compositions synthesize Western classical music with elements of various Native musical traditions.
- Ballard's classical piece *Live On, Heart of My Nation* features a Choctaw hymn.
- Ballard's classical piece *Ritmo Indio* incorporates a Sioux flageolet, a traditional Native instrument.

The student wants to emphasize a similarity between the two classical compositions. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Ballard's works synthesize Western classical music and elements of various Native musical traditions; for example, the classical piece *Live On, Heart of My Nation* features a Choctaw hymn.
- B. Ballard's composition *Live On, Heart of My Nation* features a Choctaw hymn, while *Ritmo Indio* incorporates a Sioux flageolet.
- C. Ballard's *Live On, Heart of My Nation* features a Choctaw hymn, and *Ritmo Indio* incorporates a Sioux flageolet; as such, both synthesize Western classical music with elements of Native musical traditions.
- D. Ballard's compositions synthesize Western classical music with elements of various Native musical traditions, such as a Choctaw hymn.

Question 26

While researching a topic, a student has taken the following notes:

- The A.M. Turing Award is a prestigious award given for “major contributions of lasting importance to computing.”
- Shafi Goldwasser won the award in 2012 for transformative work that laid the foundations for digital cryptography.
- Maurice Wilkes won the award in 1967 for creating the first computer with an internally stored program.

The student wants to emphasize the order in which Shafi Goldwasser and Maurice Wilkes won the A.M. Turing Award. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Shafi Goldwasser and Maurice Wilkes both won the A.M. Turing Award, which is given for “major contributions of lasting importance to computing.”
- B. It was in 1967 that Maurice Wilkes won the A.M. Turing Award.
- C. Maurice Wilkes won the A.M. Turing Award in 1967; Shafi Goldwasser won it later, in 2012.
- D. In 2012, Shafi Goldwasser won the A.M. Turing Award for transformative work that laid the foundations for digital cryptography.

Question 27

While researching a topic, a student has taken the following notes:

- The Clovis First hypothesis proposes that ancient humans known as Clovis were the first to populate North America.
- The Clovis arrived around 13,000 years ago.
- Stone tools found at Fort Rock Cave date to approximately 15,500 years ago.
- Butchered mammoth bones found at Selby date to approximately 17,000 years ago.
- These items support the idea that humans were present in North America before the Clovis population arrived.

The student wants to emphasize a similarity between the items found at Fort Rock Cave and those found at Selby. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The Clovis First hypothesis proposes that the Clovis, arriving around 13,000 years ago, were the first humans in North America.
- B. Stone tools from Fort Rock Cave date to approximately 15,500 years ago; in addition, butchered mammoth bones (which date to approximately 17,000 years ago) were discovered at Selby.
- C. Although the butchered mammoth bones found at Selby are older than the stone tools found at Fort Rock Cave, both clearly predate the arrival of the Clovis population.
- D. Objects discovered at Fort Rock Cave date to approximately 15,500 years ago; those found at Selby precede them, however, dating to approximately 17,000 years ago.

Section 1, Module 2: Reading and Writing

27 Questions

Question 1

A variety of technologies make use of magnets incorporating rare earth elements, such as gadolinium. For technology manufacturers, dependence on rare earth elements is a significant _____: such elements are extracted at large scale in only a small number of locations, making any supply chain requiring them highly fragile.

Which choice completes the text with the most logical and precise word or phrase?

- A. redundancy
- B. discrepancy
- C. vulnerability
- D. insolvency

Question 2

Asia's Gobi Desert and California's Central Valley are the main sources of dust in western North America's Sierra Nevada mountains. To investigate their relative contributions, Sarah Aciego et al. collected and recorded the isotopic signatures of dust collected at a low-elevation site (the San Joaquin Experimental Range) and two higher-elevation sites (Providence and Soaproot) in the Sierra Nevada, as well as the isotopic signatures of Gobi and Central Valley dust. Aciego et al. concluded that while the contribution of Central Valley dust, which has a lower ratio of strontium-87 to strontium-86 ($^{87}\text{Sr}/^{86}\text{Sr}$) than Asian dust does, remained mostly unchanged at the lowest-elevation site for the duration of the study, the contribution of Central Valley dust increased at the higher elevations.

Which choice best reflects the researchers' conclusions?

- A. Over the course of the study, the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of dust collected at the San Joaquin Experimental Range was relatively constant, whereas the $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of dust collected at Providence and Soaproot decreased.
- B. As the study went on, the $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of dust collected at Providence and Soaproot grew closer to the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of dust from Asia, whereas the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of dust collected at the San Joaquin Experimental Range grew closer to the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of dust collected from the Central Valley.
- C. Throughout the study, the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of dust collected at the San Joaquin Experimental Range was consistently higher than the $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of dust collected at Providence and Soaproot.
- D. By the end of the study, the $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of dust collected at Providence and the San Joaquin Experimental Range had become very similar, whereas the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of dust collected at Soaproot remained closer to the ratio of dust from Asia.

Question 3

Text 1

Isaac Asimov, author of *Nemesis* and *The End of Eternity*, is highly regarded despite his mediocre writing style. His prose is workmanlike; his characters are flat and discuss ideas rather than emotions. That his work is enjoyable despite this is a testament to his prodigious imagination—even if people read his books only for the ideas, they will have plenty to consider.

Text 2

Asimov is critiqued for his style, but it is wrong to fault a writer for failing to do what he never intended to do. For example, although most of his novel *Foundation* consists of people discussing science and politics and we find out little about his characters' lives, Asimov wanted to convey the vast sweep of human history over centuries, and one of his points is that at such a timescale, individuals don't matter. Thus his lack of characterization is central to Asimov's thematic aims.

Based on the texts, how would the author of Text 2 most likely respond to the evaluation of Asimov's writing style in the underlined claim in Text 1?

- A. By arguing that Asimov's writing style should be judged against the styles of historians rather than those of fiction writers
- B. By pointing out that the flatness of Asimov's characters is a feature of *Nemesis* and *The End of Eternity* but not of *Foundation*
- C. By agreeing that Asimov's prose is flawed but disagreeing about the quality of his characters
- D. By asserting that Asimov's minimal characterization is likely a conscious choice that helps him convey an important idea

Question 4

In 1967, archaeologist James Mellaart claimed that at its peak, Çatalhöyük—a 33-acre Neolithic village in present-day Turkey that dates from approximately 7100 to 5950 BCE—housed between 8,000 to 10,000 residents, an estimate based on the excavation of dense clusters of mud brick buildings that were thought to have been dwellings. Arguing that such estimates should be reevaluated, Ian Kuijt and Arkadiusz Marciniak (2024) cite radiocarbon data and ethnographic modeling that suggest the heterogeneity of building use in Çatalhöyük, with many individual buildings being inhabited for relatively brief periods—20 to 30 years or less—before being abandoned or repurposed for nonresidential functions (e.g., storage). Accordingly, these recent findings challenge the assumption that _____.

Which choice most logically completes the text?

- A. the buildings excavated at Çatalhöyük at the time of Mellaart's research had served a residential purpose as early as 7100 BCE.
- B. the density of excavated buildings at Çatalhöyük can serve as a proxy for how many people inhabited the village at a given time.
- C. the use of mud bricks is a reliable indicator that a structure had been used either as a dwelling or for nonresidential purposes.
- D. the number of buildings at Çatalhöyük would have been sufficient to accommodate a population that expanded from 8,000 to 10,000 inhabitants by 5950 BCE.

Question 5

Though previous studies, including work by Jacobsen et al. (2005), have documented that wood density correlates negatively with resistance to water stress in angiosperm taxa, Trueba et al. (2017) observed no such associations among New Caledonian woody angiosperms. Trueba et al. noted that while several earlier studies included a variety of habitats—wet and dry forests, savannas—and thus comprised plants with a wide range of functional trait values, the New Caledonian investigation was restricted to rainforest taxa occupying climatically similar habitats, resulting in constrained ranges of such values.

Which choice best states the main purpose of the text?

- A. To indicate that wood density and water-stress resistance correlations in angiosperms vary significantly depending on the geographic region and climate conditions studied
- B. To explain the apparent absence of a previously observed association between two functional traits in angiosperm species sampled from an ecologically homogeneous set of habitats
- C. To describe a study whose results demonstrate that a known relationship between two functional traits in angiosperms cannot be observed when sampling is restricted to homogeneous habitats
- D. To clarify why a geographically restricted study invalidated the finding that angiosperm species exhibit a trade-off between wood density and water-stress resistance

Question 6

Marketing research recognizes three distinct types of price endings: round (like \$25.00), just-below (like \$24.99), and precise (like \$24.63). A number of studies have documented significant differences in how frequently these price endings appear in different countries' marketplaces. Although these variations could be attributable to cultural factors, researchers have yet to thoroughly investigate that connection, and it remains unclear which cultural differences may be relevant or how they might interact with other socioeconomic conditions.

Which choice best states the main purpose of the text?

- A. To analyze consumer purchasing behavior across different cultures
- B. To contrast cultural and socioeconomic explanations for differences in price endings
- C. To present the finding that three different pricing strategies are used broadly across cultures
- D. To identify a research gap regarding potential cultural influences on price endings

Question 7

A light-year measures distance, not time, despite its potentially misleading name. It represents the distance that light travels through a vacuum in one Earth year—approximately 5.88 trillion miles or 9.46 trillion kilometers. Astronomers use this unit because conventional distance measures are impractical for cosmic scales. For context, Proxima Centauri, the nearest star to our solar system, is 4.24 light-years away, while the Andromeda Galaxy lies approximately 2.537 million light-years from Earth.

Which choice best describes the overall structure of the text?

- A. It highlights fundamental challenges in measuring vast cosmic distances, identifies limitations of conventional measurement units, proposes alternative approaches to conceptualizing space, and suggests how astronomical measurements might be improved in the future.
- B. It corrects a possible misconception about a scientific term, provides its precise definition with numerical equivalents, explains why the unit is used in astronomy, and offers examples of its application to cosmic objects.
- C. It contrasts various astronomical measurement systems, analyzes the advantages and limitations of each method, evaluates their accuracy for different cosmic scales, and demonstrates how they complement each other.
- D. It chronicles the historical evolution of astronomical measurement techniques, describes how earlier distance calculations were performed, explains how the light-year concept was developed by scientists, and gives two examples of distances measured in light-years.

Question 8

Humans are overwhelmingly right-hand dominant (85–90%) and can thus be said to exhibit strong population-level right-handedness (PLRH). Among studies of nonhuman primates, William C. McGrew's 1988 study of captive gorillas purported to show PLRH, while William C. McGrew and colleagues' 1999 study of wild chimpanzees did not. Overall, the studies claiming PLRH in nonhuman primates find much lower incidences of right-handedness than humans exhibit, but it's worth noting that studies of captive primates tend to show significantly greater incidences of right-handedness than studies of wild primates do, therefore raising the possibility that _____.

Which choice most logically completes the text?

- A. the apparent discrepancy between the studies' results may be partly attributable to the 1988 study using different criteria when assessing hand dominance than were used in the 1999 study.
- B. the greater exposure to humans among the captive gorillas induced them to acquire more right-handed behaviors.
- C. the number of individuals in the study of wild chimpanzees is insufficient to preclude the claim that nonhuman primates exhibit PLRH.
- D. the number of individuals in the study of captive gorillas is insufficient for a robust claim regarding evidence of PLRH.

Question 9

In *The Jazz Continuum*, LaTasha Barnes, Quynn Johnson, and other dancers seamlessly interweave recent dance styles, including breaking and waacking, with elements of the Lindy Hop, a popular swing dance of the 1930s with roots in the Black American community of Harlem, New York. Inspired by Barnes's curiosity about the similarities between the Lindy Hop and these seemingly unrelated dance styles, the show's choreography makes visible how intertwined their histories actually are. The show is therefore a testament to the Lindy Hop's enduring influence on American culture.

Which choice best states the function of the underlined portion in the text as a whole?

- A. It addresses the reasons why the historical connections between the dance styles mentioned earlier in the text have been generally overlooked.
- B. It provides evidence for the text's claim that the Lindy Hop has had a greater cultural impact than the other dances featured in *The Jazz Continuum* have had.
- C. It identifies the specific ways in which the Lindy Hop is similar to the contemporary dance styles discussed in the text.
- D. It identifies the motivation for *The Jazz Continuum*'s choreographic approach that is described earlier in the text.

Question 10

Narratives of American revolutionary history often imply that the US Declaration of Independence (published in 1776) was a document that specifically resulted from the presumably exceptional nature of the soon-to-be United States—in this interpretation, the Declaration is a singular instrument born from a response to injustice that was intended to announce the birth of a new kind of nation and win foreign allies to that nation's cause. But this interpretation neglects to place the Declaration in the larger context of eighteenth-century Western political philosophy—the Declaration was only one of several significant works published during 1776, a remarkable year for political thought.

Which statement, if true, would most strongly support the claim in the underlined portion?

- A. Newspapers in 1776 were often read aloud in public places, ensuring that even those who did not read newspapers themselves could still knowledgeably participate in political discussions.
- B. In late 1776, Benjamin Franklin traveled to France to seek French support for the American cause, eventually leading to France providing both military and financial aid to the Americans.
- C. George Campbell's *The Philosophy of Rhetoric*, a highly influential attempt to develop a foundational theory of how humans communicate, was published in 1776.
- D. In 1776, the English historian Edward Gibbon began publishing *The History of the Decline and Fall of the Roman Empire*, a work whose novel theories of empire collapse proved politically influential.

Question 11

Text 1

In parts of New Zealand, the stoat is a major predator of the house mouse. Researcher G.L. Blackwell and colleagues found that when this predation pressure on house mice was temporarily reduced, their numbers significantly increased. This finding illustrates a foundational ecological principle: predators control prey population numbers.

Text 2

François Potvin and colleagues found that excluding wolves from a site in Quebec, Canada, where they typically prey on white-tailed deer had no significant effect on white-tailed deer abundance. Many other predation-reduction studies show an increase in prey abundance, but those studies often focus on small, rapidly reproducing prey, like birds, mice, and frogs, rather than large, slowly reproducing prey, like white-tailed deer, which could account for the difference between these results and Potvin and colleagues' results.

Based on the texts, the author of Text 2 would most likely agree with which statement about the “principle” mentioned in Text 1?

- A. It has been challenged by some studies, but the findings of those studies have not been widely accepted.
- B. It is plausible, but many of the studies that support it have methodological flaws.
- C. It has some evidential support, but it should not be regarded as universally applicable.
- D. It may be true for some predators but only because those predators share certain physical characteristics.

Question 12

O'Donoghue et al. (2021) analyzed temperature data of Jupiter's upper atmosphere observed in 2016 and 2017. When superimposed on a map of Jupiter, time-lapse data representing several hours of observations from one day in 2016 and one day in 2017 indicate an overall temperature gradient from the poles to the equator, with highest temperatures occurring at the poles, which undergo auroral heating. But the 2017 data exhibit a large oval of unusually elevated temperature separated from the aurora at one pole by a band of cooler atmosphere. No other known heating mechanism on Jupiter could account for the formation of this anomalous structure, so one hypothesis considered was that it was ejected from the auroral region toward the equator.

Which finding about Jupiter's upper atmosphere, if true, would most strongly support the hypothesis as described in the text?

- A. Comparing the data observed in 2016 with those observed in 2017 suggests that the anomalous oval of elevated temperature began forming in 2016.
- B. According to the data, auroral temperatures in 2016 were higher than those in 2017, although in both years the highest observed temperatures were harbored in auroral zones.
- C. The 2017 data viewed in time-lapse appear to show the oval of elevated temperature drifting slightly away from the area of auroral heating over a period of several hours.
- D. The 2017 data viewed in time-lapse show greater disparities between polar and equatorial temperatures at the beginning of the observation period than at the end of the period observed.

Question 13

The advent of online streaming has led many music listeners to drift away from ownership of music (through downloads or through physical media such as compact discs) and toward the streaming services iHeartRadio and Tidal, among others. Datta et al. studied the impact of this change on the variety of music that listeners consume. The researchers reasoned that the ownership model of music assigns a cost per song to acquiring a variety of music, while streaming services typically charge a flat fee for access to an entire music catalog, making variety free, which suggests that _____.

Which choice most logically completes the text?

- A. listeners who use streaming services would be more likely to give physical copies of music as gifts to others than to purchase physical copies for themselves.
- B. music publishers who choose to forgo releasing music on physical media are likely to see no change in revenue.
- C. the music choices of listeners who use streaming services would likely be more varied than those of listeners who do not use streaming services.
- D. listeners who prefer to purchase compact discs rather than use a service such as iHeartRadio or Tidal would tend to listen to older music.

Question 14

Text 1:

For decades, ornithologists assumed that if they saw a singing Bell's vireo—a bird species found in temperate North America—they must be observing a male trying to attract a mate or claim territory. As Peter J.B. Slater and Nigel I. Mann have emphasized, however, a similar assumption can't be made about birds in the tropics, where females sing as often as males do. Slater and Mann call for more research on this discrepancy between tropical and temperate female birdsong.

Text 2:

Recent evidence shows that a female Bell's vireo is as capable of song as a male is. In fact, female birdsong is more common among temperate species than currently assumed, claim Evangeline Rose and colleagues. These female songbirds sing less frequently than males do, and in duller tones, making it “easy for researchers to miss the quiet and hidden females and focus on the loud and colorful males,” says Rose.

Based on the texts, how would Rose and colleagues (Text 2) most likely respond to the assertion by Slater and Mann (Text 1) about the different prevalence of female birdsong in temperate and tropical areas?

- A. They would raise the possibility that the difference in prevalence may be due to differences in the timing of the mating season among temperate and tropical bird species.
- B. They would argue that the apparent difference in prevalence may partly reflect a difference in the ease with which female birdsong and male birdsong can be detected.
- C. They would caution that the seeming difference in prevalence may be an artifact of researchers' tendency to study birdsong among temperate species more frequently than among tropical species.
- D. They would concede that the geographic difference in prevalence is real but argue that the frequency with which male tropical birds sing has been overstated by previous researchers.

Question 15

A film studies student is researching early 20th-century film serials, which consisted of individual episodes of a single long story that were shown weekly in theaters. *The Shadow of the Eagle* is a 1932 serial that, over its 12 episodes, kept its audience interested with the suspense and drama that are typical of the crime genre. The student, however, claims that ultimately audiences of the time preferred resolution and closure over ongoing tension.

Which finding, if true, would most directly support the student's claim?

- A. Audiences of the time considered *The Shadow of the Eagle* to belong to a genre other than the crime genre.
- B. The 12th episode of *The Shadow of the Eagle* was the most expensive episode of the series to produce.
- C. Modern critics generally regard the first episode as the best installment of *The Shadow of the Eagle*.
- D. The 12th episode of *The Shadow of the Eagle* was viewed by more people than was any previous episode in the series.

Question 16

A student is writing an essay about how writers of the late 1700s responded to the French Revolution and wants to include two pieces of textual evidence that show closely aligned attitudes toward the revolution. The student has selected the following quotation, adapted from Helen Maria Williams's 1791 poem "A Farewell, for Two Years, to England":

In new-born majesty [France] seems to rise,
While sudden from the land oppression flies.
So, at the solemn hour of Nature's birth,
When brooding darkness veil'd the beauteous earth,
Heaven's awful mandate pierc'd the solid night,
"Let there be light," it said, "and there was light!"

Which quotation from another prominent writer of the same period would be most effective for the student to include along with the quotation from Williams's poem?

- A. "The age of chivalry is gone. That of sophisters, economists, and calculators has succeeded; and the glory of Europe is extinguished forever." (Edmund Burke)
- B. "Let human nature, conscious of its own infirmities, humbly solicit the protection of Omnipotence against the arrogance of human Reason, whenever, proud of her own supremacy, she presumes to pass the interdicted bounds prescribed to her finite powers." (Jane West)
- C. "[Liberty] scorns to disdain / Her righteous cause with carnage, and resigns / To Fraud and Anarchy the infuriate crowd." (Charlotte Smith)
- D. "Reason has, at last, shown her captivating face, beaming with benevolence; and it will be impossible for the dark hand of despotism again to obscure its radiance." (Mary Wollstonecraft)

Question 17

High-reward flowers that produce nectar with relatively high sugar concentrations are thought to be especially attractive to scaly-breasted hummingbirds and other hummingbird species. But when María A. Maglianesí and team monitored hummingbird visits to flowering plants, including *Heliconia irrasa*, at Los Cusingos Bird Sanctuary and other sites in Costa Rica, they found that visit frequencies weren't influenced by nectar traits. Rather, physical compatibility of flower corollas (collections of petals) and hummingbird bills more strongly predicted hummingbird-plant interaction patterns.

Which statement about hummingbird preferences is most strongly supported by the text?

- A. Hummingbirds' preferences are primarily driven by trait-matching between their bills and flower corollas, though nectar qualities are a secondary factor driving their preferences.
- B. Hummingbirds consistently prefer plant species whose flowers have physical traits that are compatible with their bills but offer relatively low nectar rewards.
- C. A floral trait found to influence which plant species hummingbirds prefer may result in certain hummingbird species visiting flowers with relatively low nectar rewards over those with higher nectar rewards.
- D. Physical compatibility between flower corollas and hummingbird bills becomes an important factor that influences hummingbird preferences when hummingbirds are uncertain about which plants produce high-reward nectar.

Question 18

Text 1

It's clear that worldwide cooperation is needed to tackle global issues such as climate change. However, it hasn't been clear how to foster strong group cohesion to motivate prosocial behavior at a global level given that humans tend to align more locally, from family to nationality. Evidence now shows that being prompted to consider the shared biology of all humans can cause individuals to identify more strongly with humanity overall, suggesting that emphasis on the "global family tree" may promote collective action.

Text 2

Perhaps counterintuitively, a focus on group distinctions can help ensure that responsibility for issues with global effects (e.g., natural resource depletion and climate change) is shared globally. For example, though young people don't always view themselves as essential parties in such matters, reminders of their collective role as future inheritors of problems left unsolved can spark their engagement.

Which choice best describes a notable difference in focus between Text 1 and Text 2?

- A. Both texts imply that global issues are equally relevant to all people living today, but Text 1 focuses on the idea that such issues will remain broadly relevant for the foreseeable future and Text 2 focuses on the idea that certain issues may soon become less relevant.
- B. Both texts establish that people need to be made aware of the specific ways global issues are likely to affect them individually, but Text 1 focuses on the empathy those details inspire and Text 2 focuses on the obligation those details entail.
- C. Both texts suggest that embracing commonalities can be beneficial when it comes to urging people to collective action to address global issues, but Text 1 focuses on the broadest scope of connection and Text 2 focuses on connections more limited in scope.
- D. Both texts indicate that global issues should be addressed with large-scale shared efforts, but Text 1 focuses on the idea that it may be most effective to link global and local coordination and Text 2 focuses on the idea that local efforts sometimes divert attention from broader coordination.

Question 19

The *Lex Licinia sumptuaria* of 103 BCE was one in a succession of ancient Roman sumptuary laws governing expenditures on luxury, primarily regulating ceremonial processions and other lavish events. Such laws, enacted against a backdrop of anxiety about social dynamism, were intended to limit aspirational overreaching for status but were infrequently enforced. Moreover, enforcement was generally limited to fines, which, when levied against high-profile offenders, were widely announced. The affluent accepted fines as a tax on privilege, even as the amounts increased with frequent revisions to the laws. Historians have therefore argued that enforcing sumptuary laws only minimally may have been a way to avoid undermining the laws' aims, given that _____.

Which choice most logically completes the text?

- A. the public announcement of fines could signal offenders' wealth and status, placing enforcement of the laws in conflict with their intent.
- B. the few fines that were imposed under the laws were tolerable to wealthy offenders and thus did not help the laws serve their intended purpose.
- C. authorities focused enforcement on the wealthiest offenders, recognizing that pursuing offenders who were unable to pay the fines could weaken fines' effectiveness as a deterrent.
- D. less-affluent citizens were reluctant to risk incurring a fine and thus voluntarily complied with the laws even in the absence of regular enforcement.

Question 20

In environments ranging from shallow tidal areas to the deep ocean, single-celled marine foraminifera have been found to contain phosphate at concentrations many times greater than those in the surrounding water. Notably, though, species from oxygen-rich environments like the Mid-Atlantic Ridge show lower phosphate concentrations than species from oxygen-poor areas. Scientists have thus suggested that especially high internal phosphate concentrations enable foraminifera to offset the effects of low oxygen.

Which choice best states the main idea of the text?

- A. There is a correlation between habitat depth and phosphate concentration in marine foraminifera, with deeper-dwelling species generally maintaining higher internal concentrations.
- B. While marine foraminifera in general have high phosphate concentrations, those concentrations tend to vary by species' habitat, possibly reflecting adaptations to different levels of oxygen.
- C. Marine foraminifera maintain similar phosphate concentrations across different ocean environments, suggesting that while this trait plays a role in offsetting low oxygen levels, that is not its only function.
- D. Marine foraminifera counterbalance the effects of reduced oxygen levels by storing phosphate at high concentrations, matching or in some cases even slightly exceeding the phosphate concentration in the surrounding seawater.

Question 21

In an extensive review of existing literature, Léna de Framond and team cataloged the prevalence of broken-wing display—a defensive behavior observed in *Haematocephalus palliatus* (American oystercatcher) and many other species—throughout the Aves class. Documentation of the display in 285 species across 52 families suggests the behavior likely evolved independently multiple times, prompting the team to consider ecological and life-history characteristics with hypothesized associations to the behavior's emergence, including traits related to reproduction investment and future reproduction potential. Based on their review of those traits, the team concluded that _____.

Scientific name	Common name	Broken-wing display?	Incubation duration (days)	Parental incubators
Setophaga caeruleus	black-throated blue warbler	Yes	13	1 parent
Tetrastes sewerzowi	Chinese grouse	No	29	1 parent
Haematopus ostralegus	Eurasian oystercatcher	No	28	2 parents
Zenaida macroura	mourning dove	Yes	15	2 parents

Which choice most effectively uses data from the table to complete the conclusion?

- A. broken-wing display is most often observed in species with less opportunity to reproduce in a year due to longer incubation periods.
- B. incubation duration and capacity for multiple broods are more strongly associated with the use of broken-wing display than the number of parental incubators is.
- C. capacity for multiple broods, number of parental incubators, and incubation duration are equally associated with the use of broken-wing display.
- D. among species with more than one parental incubator, the use of broken-wing display is associated with greater incubation duration.

Question 22

Strontium Isotope Ratios for an Igneous Rock Sample

Mineral formation stage	$^{87}\text{Sr}/^{86}\text{Sr}$ ratio
Crystal core	0.71685
Crystal rim	0.71548
Crystal matrix	0.71416

A student is studying igneous rock formations on Elba Island, Italy, and reviews published data on the ratio of strontium isotopes ($^{87}\text{Sr}/^{86}\text{Sr}$) in minerals from a specific rock sample. These isotope ratios provide information about the composition of the magma that formed the rock. Ratios below 0.710 suggest more material from the mantle in the magma, while ratios increasing above that number usually indicate greater proportions of material from the continental crust. The minerals in the rock sample record three growth stages: crystal core zones formed first, then rim zones, and finally the surrounding matrix. The student concludes that the rock sample _____.

Based on the information given, which choice most effectively uses data from the table to complete the student's conclusion?

- A. shows evidence of increasing continental crust material in later-forming minerals.
- B. contains $^{87}\text{Sr}/^{86}\text{Sr}$ ratios too similar to provide reliable information about magma sources.
- C. does not show the expected $^{87}\text{Sr}/^{86}\text{Sr}$ ratios across the stages of mineral formation.
- D. formed from magma with decreasing but persistent crustal influence.

Question 23

Highly Traveled Commercial Airline Routes in 2017–18

Location 1	Location 2	Distance (km)	2018 passengers	2017 passengers	Type
Hong Kong	Seoul-Incheon	2,080	3,081,942	3,198,132	international
Beijing	Shanghai-Hongqiao	1,081	6,518,997	6,833,684	domestic
Hong Kong	Bangkok-Suvarnabhumi	1,694	3,490,988	3,438,628	international
Mexico City	Monterrey	729	3,474,971	3,202,091	domestic

A study of commercial airline routes that carried the greatest number of passengers in 2017 and 2018 shows that, in general, domestic routes (routes within a single country) carried many more passengers than international routes. However, a few international routes were so popular that they carried more travelers than some of the most highly traveled domestic routes, as can be seen when comparing the _____.

Which choice most effectively uses data from the table to complete the example?

- A. distance between Mexico City and Monterrey with the distance between Hong Kong and Bangkok-Suvarnabhumi.
- B. number of passengers who traveled between Beijing and Shanghai-Hongqiao in 2017 with the number who traveled between Hong Kong and Seoul-Incheon during that same year.
- C. number of passengers who traveled between Hong Kong and Bangkok-Suvarnabhumi in 2018 with the number who traveled between Mexico City and Monterrey during that same year.
- D. number of passengers who traveled between Hong Kong and Bangkok-Suvarnabhumi in 2017 with the number who traveled that same route in 2018.

Question 24

Examples of Hoards Found in Scotland and Wales

Hoard name	Date of contents	Year of discovery	Description
Glenmore Hoard	17th century CE	2015	gold coins
Ashford Hoard	24th–19th century BCE	1938	gold pieces
Blackwater Hoard	1st century BCE	1887	gold pieces

Deposits of valuable objects, called hoards, have been unearthed in many different parts of Scotland and Wales. Technological advancements in the twenty-first century have made it easier to locate hoards, but plenty of hoards were found earlier. For example, _____

Which choice most effectively uses data from the table to complete the statement?

- A. the Blackwater Hoard and Ashford Hoard were both found before 2000.
- B. the Blackwater Hoard, Ashford Hoard, and Glenmore Hoard have all been found since 1887.
- C. the Ashford Hoard was found in 1887.
- D. Ashford Hoard and Glenmore Hoard were both discovered in the 1900s.

Question 25

Physicists like Kathryn Zurek have begun to consider that dark matter might not be one single particle _____ such an expanded theoretical framework, an entire hidden world of dark matter particles—with unique forces and dynamics of their own—could exist parallel to our own.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. type, which in
- B. type. In
- C. type, in
- D. type in

Question 26

With evidence of long-term use over centuries (a pattern rarely seen in early Caribbean sites) _____ conventional theories about highly mobile foraging groups, the Orbit site is an example of what archaeologists call persistent placemaking.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. challenging
- B. have challenged
- C. challenged
- D. has challenged

Question 27

In biology there is no single agreed-upon concept of what defines a species; _____ numerous competing species concepts are used within the field, a dilemma known as the species problem. The biological species concept, described in 1942 by Mayr, defines species in accordance with reproductive isolation, while the phylogenetic species concept, delineated by Queiroz and Donoghue in 1988, organizes species according to monophyletic groups.

Which choice completes the text with the most logical transition?

- A. in addition,
- B. for instance,
- C. as a result,
- D. however,

Section 2: Math

Section 2, Module 1: Math

1 SAT Math

Which expression is equivalent to $240x^2 + 56$?

- A. $308x^2 + 1$
- B. $78x^2 + 30$
- C. $8(30x^2 + 7)$
- D. $296(x^2 + 1)$

2 SAT Math

$$x^2 + 22x = -121$$

What is the solution to the given equation?

- A. -22
- B. -121
- C. 11
- D. -11

3 SAT Math

Kaden owns a business that makes signs. The equation $t = 18s + 15$ gives the total time t in minutes, it takes Kaden's business to make s signs. How long, in minutes, does it take Kaden's business to make 100 signs?

- A. 3,300
- B. 1,518
- C. 1,815
- D. 1,800

4 SAT Math

In the ancient Roman measurement system, a leuga was a unit of length equal to 7,500 pedes. In this measurement system, how many pedes were equivalent to 2.5 leugas?

5 SAT Math

The length of each side of a square is 68 inches. Which expression gives the area of the square?

- A. $68/68$ square inches
- B. $68 + 68$ square inches
- C. 68^2 square inches
- D. $180(68)$ square inches

6**SAT Math**

$$3|x| = 45$$

Which value is a solution to the given equation?

- A.** 42
- B.** 15
- C.** 48
- D.** 135

7**SAT Math**

At a particular restaurant, the total cost of 6 chicken wings and a salad is 15.30 dollars. If the cost of each chicken wing is 1.25 dollars and there is no sales tax, what is the cost, in dollars, of the salad?

- A.** 7.50
- B.** 14.05
- C.** 1.25
- D.** 7.80

$$x > 55$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
50	0
51	0
52	0

B.

x	y
56	0
57	0
58	0

C.

x	y
45	0
46	0
47	0

D.

x	y
35	0
36	0
37	0

9 SAT Math

$$|x - 9| = 14$$

What is the sum of the solutions to the given equation?

- A. 23
- B. 18
- C. 28
- D. 0

10 SAT Math

The population of a certain city doubled every 60 years from 1742 to 1982. The population of this city was 320,000 in 1982. What was the population of this city in 1742?

11 SAT Math

Which expression is equivalent to $(2x^3+3x-9)-(7x^2+8x+4)$?

- A. $-5x^3+11x-5$
- B. $-5x^3+5x-13$
- C. $2x^3-7x^2-5x-13$
- D. $2x^3-7x^2+11x-5$

12 SAT Math

$$7x - 4y = 5$$

$$2x + y = 7$$

The solution to the given system of equations is (x, y) . What is the value of $x + y$?

13 SAT Math

The side lengths of right triangle RST are $RS = 360$, $ST = 224$, and $TR = 424$. Triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. What is the value of $\tan W$?

A. $28/53$

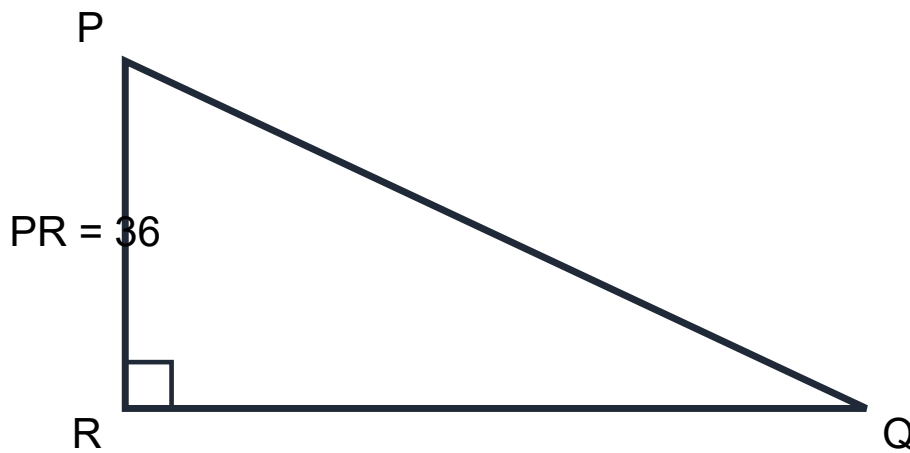
B. $28/45$

C. $45/53$

D. $45/28$

14 SAT Math

In right triangle PQR, angle R is a right angle, $PR = 36$, and $\sin Q = 1/2$. What is the length of PQ?



Note: Figure not drawn to scale.

15 SAT Math

A baker has a total of 11 cups of flour and enough other ingredients to make banana pancakes and apple pancakes. The baker needs 1.5 cups of flour for a batch of banana pancakes and 0.75 cup of flour for a batch of apple pancakes. If the baker makes 5 batches of banana pancakes, what is the greatest number of whole batches of apple pancakes the baker can make?

16 SAT Math

$g(x)=9-(5x-1)^2$ The function g is defined by the given equation. For what value of x does $g(x)$ reach its maximum?

- A. $-(1)/(5)$
- B. $-(1)/(9)$
- C. $(1)/(9)$
- D. $(1)/(5)$

17 SAT Math

A circle has center O , and points P and Q lie on the circle such that segment PQ is a diameter of the circle. The length of arc PQ is 52 centimeters. What is the circumference, in centimeters, of the circle?

- A. 26
- B. 52
- C. 104
- D. 360

18 SAT Math

A proposal for a construction project was included on a city election ballot. A TV news report stated that 3 times as many people voted in favor of the proposal as people who voted against it. An internet article reported that 12,000 more people voted in favor of the proposal than voted against it. Based on these data, how many people voted against the proposal?

- A. 6,000
- B. 12,000
- C. 18,000
- D. 36,000

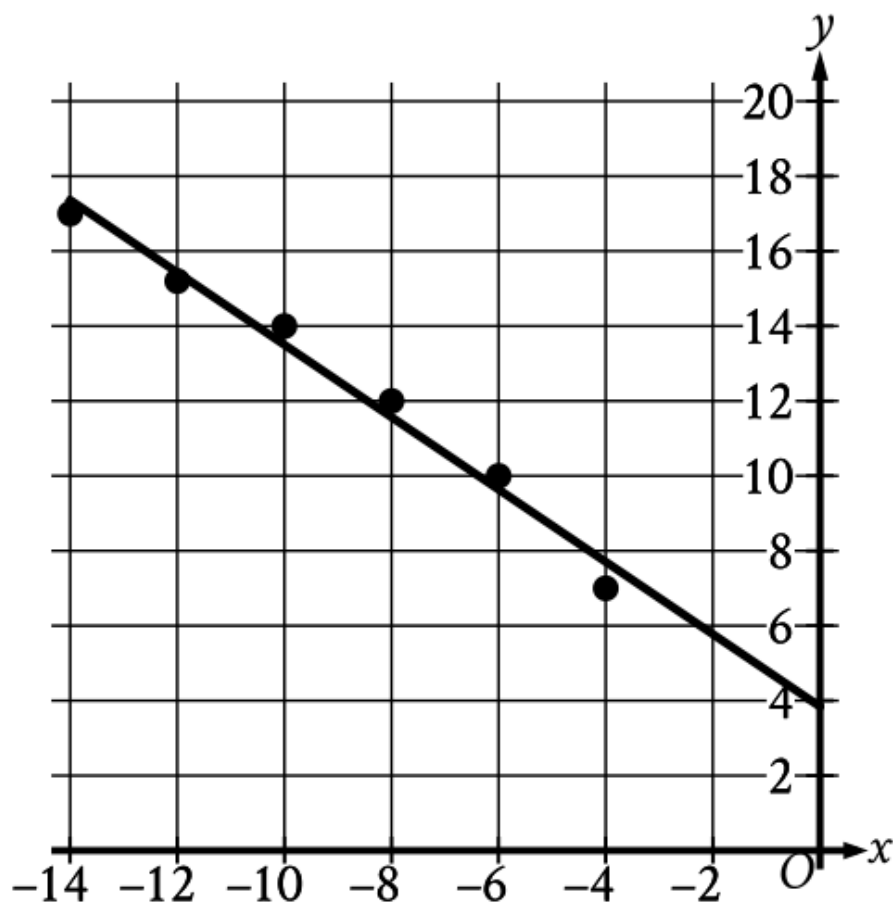
19 SAT Math

If $\sqrt{x-6}=7$, what is the value of $(x-6)$?

- A. 7
- B. 43
- C. 49
- D. 98

20 SAT Math

In the xy -plane, line k has a slope of 4 and a y -intercept of $(0, -28)$. What is the x -coordinate of the x -intercept of line k ?



Which equation best represents the line of best fit shown?

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown. Which of the following equations best represents the line of best fit shown?

- A. $y = x - 17.4$
- B. $y = -x + 17.4$
- C. $y = x - 3.8$

D. $y = -x + 3.8$

22 SAT Math

At a conference hall, there are a total of 300 visitors. Each visitor is located in either room A, room B, or room C. If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.42 and the probability of selecting a visitor who is located in room B is 0.18. How many visitors are located in room C?

Section 2, Module 2: Math

1 SAT Math

$$w = 19$$

$$1 = (w - 19)^2/25 + (p + 3)^2/49$$

A solution to the given system of equations is (p, w) . What is one possible value of p ?

- A. -3
- B. 4
- C. 5
- D. 7.01

2 SAT Math

In a certain region, the population of *Phasianus colchicus* is estimated each year by counting the number of these birds observed along certain roads in the region. From 1990 to 2005, the number of these birds counted per mile of road increased by an average of 3.4% each year over the number counted per mile of road in the previous year. Based on this average, if 3.22 *P. colchicus* were counted per mile of road in the region in 1990, which of the following best approximates the number counted per mile of road in 2005?

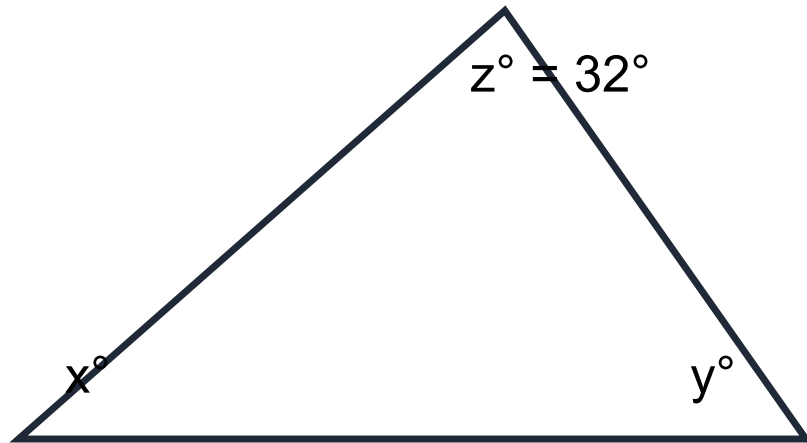
- A. $1.034(0.966)^{15}$
- B. $3.22(0.966)^{15}$
- C. $3.22(1.034)^{15}$
- D. $0.966(3.22)^{15}$

A researcher agitated a quantity of milk inside a graduated beaker, which produced foam on the surface of the milk. The predicted height, in centimeters (cm), of the foam can be modeled by the function $f(t) = 3.90(0.86)^t$, where t is the time, in minutes, after the milk was agitated. Which of the following is the best interpretation of the statement “ $f(0.40)$ is approximately 3.7” in this context?

- A.** 0.40 minutes after the milk was agitated, the predicted height of the foam is approximately 3.7 cm.
- B.** 0.40 minutes after the milk was agitated, the predicted height of the foam has decreased by approximately 3.7 cm.
- C.** 3.7 minutes after the milk was agitated, the predicted height of the foam is approximately 0.40 cm.
- D.** 3.7 minutes after the milk was agitated, the predicted height of the foam has decreased by approximately 0.40 cm.

4 SAT Math

In the triangle shown, $z = 32^\circ$. Which of the following represents y in terms of x ?



Note: Figure not drawn to scale.

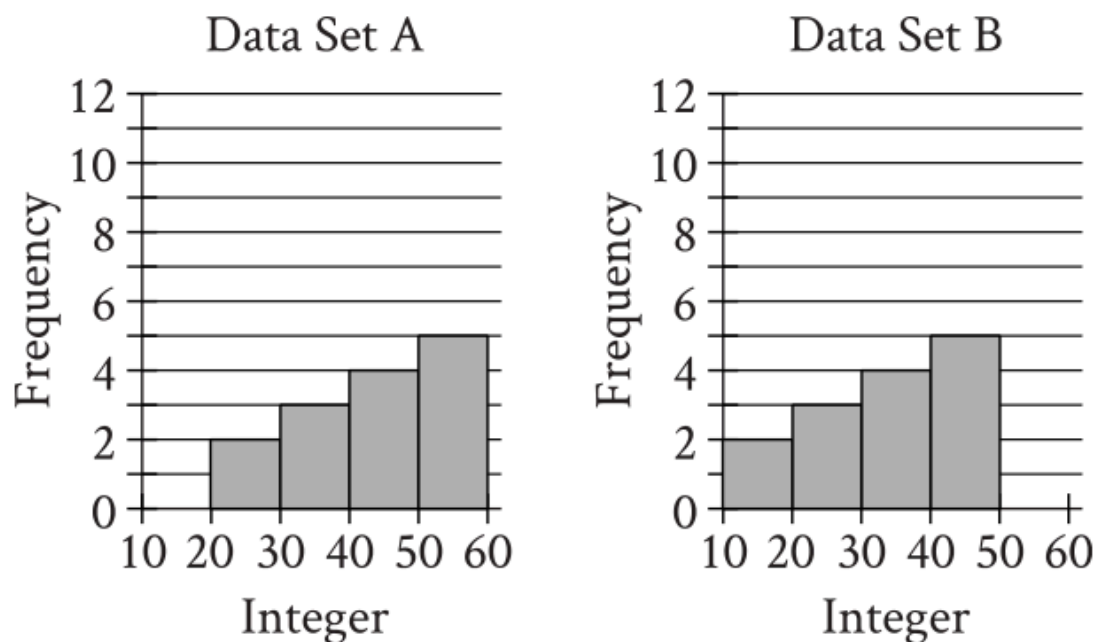
- A. $y = 32 + x$
- B. $y = 58 + x$
- C. $y = 212 - x$
- D. $y = 148 - x$

5 SAT Math

What is the sum of the solutions to the given equation? $4x^2 - 6x - 17 = 0$

A scientist studied the effects of mixing two chemicals, chemical A and chemical B, by gradually adding more of each chemical to a mixture of the two chemicals. At the start of the study, the mass of chemical A in the mixture was equal to the mass of chemical B in the mixture. From the start of the study to the end of the study, the mass of chemical A in the mixture increased by 2200%, and the mass of chemical B in the mixture increased by 330%. At the end of the study, approximately how many times greater was the mass of chemical A in the mixture than the mass of chemical B in the mixture?

- A. 6.67
- B. 5.35
- C. 25.30
- D. 18.70



4 integers each are summarized in the histograms shown. For the first interval represents the frequency of integers greater than or equal to 10, but less than 20. The second interval represents the frequency of integers greater than or equal to 20, but less than 30, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

Two data sets of 14 integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to 10, but less than 20. The second interval represents the frequency of integers greater than or equal to 20, but less than 30, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

- A. 0
- B. 1
- C. 10
- D. 14

8 SAT Math

$$\left(\frac{1}{3}\right)x^2 + 58 = t + 4$$

In the given equation, t is a constant. The equation has exactly one real solution. What is the value of t ?

- A. 58
- B. -174
- C. 54
- D. -54

9 SAT Math

$t(v - 89) + t = w$ The given equation relates the positive numbers w , t , and v . Which equation correctly expresses v in terms of t and w ?

- A. $v = w/t + 88$
- B. $v = w/t - 89$
- C. $v = (w - t)/t + 88$
- D. $v = (w - t)/t - 89$

10 SAT Math

The height of a right circular cylinder is 31 inches, and the circumference of its base is 310 inches. Which expression represents the total surface area, in square inches, of the cylinder?

- A. $(31)(310)$
- B. $\pi((155)/(\pi))^2(31)$
- C. $(31)(310) + \pi((155)/(\pi))^2$
- D. $(31)(310) + 2\pi((155)/(\pi))^2$

11 SAT Math

In 2005, Aster earned 14% more than in 2004, and in 2006 Aster earned 4% more than in 2005. If Aster earned y times as much in 2004 as in 2006, which of the following is closest to the value of y ?

- A. 0.2857
- B. 0.5600
- C. 0.8435
- D. 1.1856

12 SAT Math

$f(x) = (2x - 12)(2x + 16)$ What is the minimum value of the given function?

13 SAT Math

A rectangle is inscribed in a circle such that the diagonal of the rectangle is twice the length of the shortest side of the rectangle. The area of the rectangle is $961\sqrt{3}$ square units. What is the length, in units, of the radius of the circle?

14 SAT Math

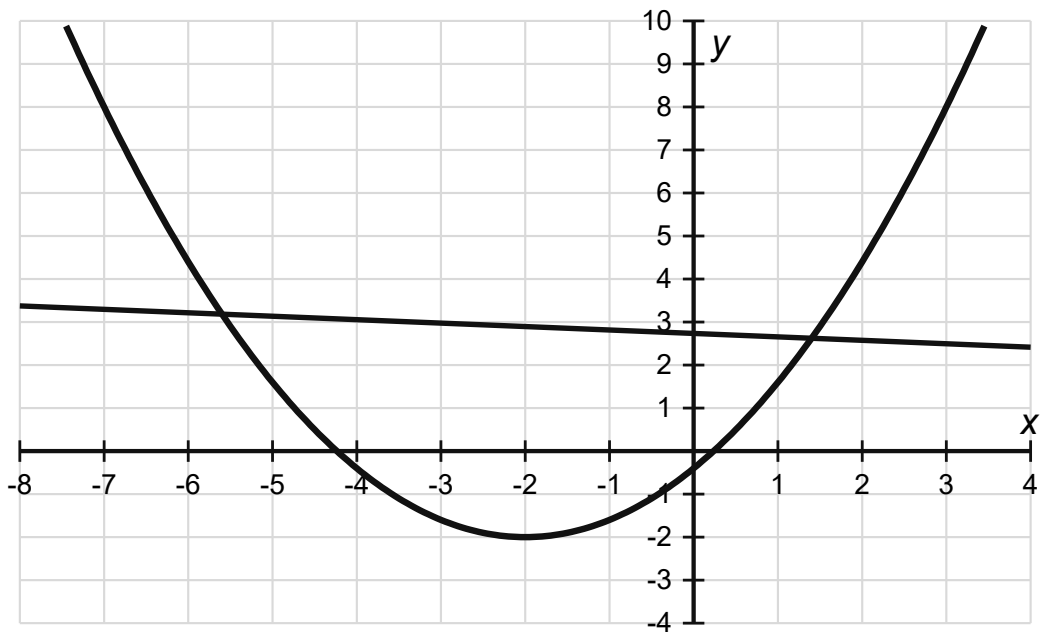
From a population of 30,000 people, 3,000 were chosen at random and surveyed about a proposed piece of legislation. Based on the survey, it is estimated that 70% of people in the population support the legislation, with an associated margin of error of 1.6%.

Based on these results, which of the following is a plausible value for the total number of people in the population who support the proposed legislation?

- A.** 900
- B.** 2,100
- C.** 8,760
- D.** 20,760

15 SAT Math

The graph of a system of a linear equation and a quadratic equation is shown. A solution to the system is (x, y) . Which of the following could be the value of x ?



- A. -6
- B. -5.6
- C. -3
- D. -2

16 SAT Math

An initial mixture consists of 300 grams of metal X and 20 grams of metal Y. During a lab experiment, for each 8 grams of metal X that are added to the initial mixture, 16 grams of metal Y are also added to the initial mixture. For the resulting mixture, which of the following equations gives y , the total mass of metal Y, in grams, when the total mass of metal X is x grams?

- A. $y = 2x + 20$
- B. $y = 35x + 20$
- C. $y = 4x - 292$
- D. $y = 2x - 580$

17 SAT Math

To cut a lawn, Jamal charges a fee of \$18.00 for his equipment and \$8.75 per hour spent cutting a lawn. Sofia charges a fee of \$10.00 for her equipment and \$10.75 per hour spent cutting a lawn. If x represents the number of hours spent cutting a lawn, what are all the values of x for which Sofia's total charge is greater than Jamal's total charge?

- A. $x < 3$
- B. $4 \leq x \leq 5$
- C. $x > 4$
- D. $3 \leq x \leq 4$

18 SAT Math

$$x^2 + 31x + 1 = 0$$

What is one of the solutions to the given equation?

- A. $(-31 + \sqrt{(31^2 + 4)})/2$
- B. $(-31 + \sqrt{(31^2 - 4)})/2$
- C. $(31 + \sqrt{(31^2 - 4)})/2$
- D. $(-1 + \sqrt{(31^2 - 4)})/2$

19 SAT Math

$$x^2 + 16x + y^2 = 12y + 101$$

In the xy -plane, the graph of the given equation is a circle. What is the length of the circle's diameter?

- A. $2\sqrt{201}$
- B. $\sqrt{201}$
- C. $\sqrt{402}$
- D. $2\sqrt{101}$

20 SAT Math

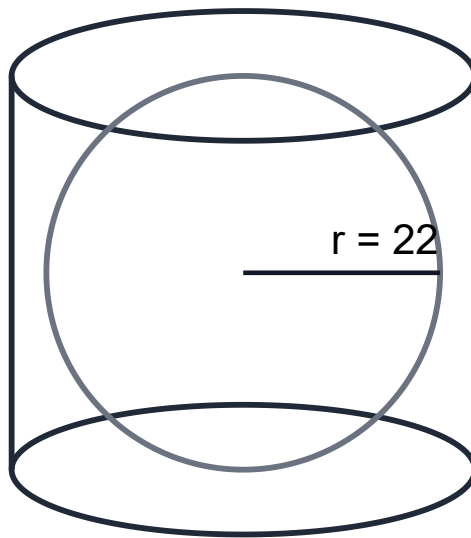
$x + 33 = \sqrt{(x + 45)} - 12$ The solutions to the given equation are s and t , where $s < t$. What is the value of s ?

21 SAT Math

In right triangle ABC , angle C is the right angle and $BC = 17$. Point D lies on side AB and point E lies on side AC such that DE is parallel to BC and $CE = 2AE$. What is the length of DE ?

22 SAT Math

A sphere is inscribed in a right circular cylinder, as shown. The sphere touches the center of each base of the cylinder. The sphere and the cylinder have the same radius r , where $r = 22$ centimeters. What is the surface area, in square centimeters, of the cylinder?



Note: Figure not drawn to scale.

- A. $1,936\pi$
- B. $2,904\pi$
- C. $5,808\pi$
- D. $21,296\pi$

1	C	1	C	1	C	1	B
2	B	2	A	2	D	2	C
3	D	3	D	3	C	3	A
4	C	4	B	4	18750	4	D
5	D	5	B	5	C	5	3/2
6	C	6	D	6	B	6	B
7	D	7	B	7	D	7	B
8	C	8	B	8	B	8	C
9	A	9	D	9	B	9	A
10	B	10	D	10	20000	10	D
11	D	11	C	11	C	11	C
12	C	12	C	12	4.8	12	-196
13	C	13	C	13	D	13	31
14	B	14	B	14	72	14	D
15	B	15	D	15	4	15	B
16	A	16	D	16	D	16	D
17	A	17	C	17	C	17	C
18	C	18	C	18	A	18	B
19	C	19	A	19	C	19	A
20	D	20	B	20	7	20	-45
21	C	21	B	21	D	21	17/3
22	C	22	D	22	120	22	B
23	B	23	C				
24	A	24	A				
25	C	25	B				
26	C	26	A				
27	C	27	C				